

University of California, Davis
 Managerial Economics (ARE) 106, Fall 2025
 Instructor: John H. Constantine
Discussion Section 4

Problem 1: (text 4.12)

The data for this problem are (i) y = pounds of chicken consumed and (ii) x = price per pound. There are 52 observations in this data set.

In this problem we will consider 4 functional forms:

- class {
1. Linear: $q = \beta_1 + \beta_2 p$ functional form we have focused on to date.
 2. Lin-log: $q = \beta_1 + \beta_2 \ln(p)$
 3. Log-lin: $\ln(q) = \beta_1 + \beta_2 p$
 4. Log-log: $\ln(q) = \beta_1 + \beta_2 \ln(p)$
 5. Inverse: $q = \beta_1 + \beta_2 \left(\frac{1}{p}\right)$

This problem is to introduce different functional forms.

In addition, the first three data points (of this set of 52 data points).

TABLE 9-11 SUMMARY OF FUNCTIONAL FORMS.

Model	Form	Slope = $\frac{dY}{dX}$	Elasticity = $\frac{dY}{dX} \cdot \frac{X}{Y}$
Linear	$Y = B_1 + B_2 X$	B_2	$B_2 \left(\frac{X}{Y}\right)^*$
Log-linear	$\ln Y = B_1 + B_2 \ln X$	$B_2 \left(\frac{Y}{X}\right)$	B_2
Log-lin	$\ln Y = B_1 + B_2 X$	$B_2(Y)$	$B_2(X)^*$
Lin-log	$Y = B_1 + B_2 \ln X$	$B_2 \left(\frac{1}{X}\right)$	$B_2 \left(\frac{1}{Y}\right)^*$
Reciprocal	$Y = B_1 + B_2 \left(\frac{1}{X}\right)$	$-B_2 \left(\frac{1}{X^2}\right)$	$-B_2 \left(\frac{1}{XY}\right)^*$
Log-inverse	$\ln(Y) = B_1 - B_2 \left(\frac{1}{X}\right)$	$B_2 \left(\frac{Y}{X^2}\right)$	$B_2 \left(\frac{1}{X}\right)$

Note: * Indicates that the elasticity coefficient is variable, depending on the value taken by X or Y or both. When no X and Y are specified, in practice, these elasticities are often measured at the mean values $\bar{X}$ and $\bar{Y}$.

Given {

Problem 1:

Refer to the computer printouts for the four problems below.

- State your regression results from the “Linear” or “Lin-Lin” regression model.
- State your regression results from the “Lin-Log” regression model.
- State your regression results from the “Log-Lin” regression model.
- State your regression results from the “Log-Log” or “Log-Linear” regression model.
- State your regression results from the “Inverse” or “Reciprocal” regression model.

Ignore the t-stats as all are statistically significant.

For each functional form above, calculate the elasticity for E_{qp} when evaluated at $P = 1.31$.

1 LIN-LIN

Regression Statistics	
Multiple R	0.842929968
R Square	0.71053093
Adjusted R Square	0.704741549
Standard Error	6.512777431
Observations	52

$$q = f(p)$$

ANOVA

	df	SS	MS	F	Significance F
Regression	1	5205.749912	5205.749912	122.7300262	4.58359E-15
Residual	50	2120.813493	42.41626986		
Total	51	7326.563405			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	57.95564074	2.575223786	22.50508909	8.08978E-28	52.78315154	63.12812994	52.78315154	63.12812994
p	-18.40283001	1.661151341	-11.07835846	4.58359E-15	-21.73935068	-15.06630935	-21.73935068	-15.06630935

2 LIN-LOG

Regression Statistics	
Multiple R	0.902115815
R Square	0.813812945
Adjusted R Square	0.810089203
Standard Error	5.22323897
Observations	52

$$q = f(\ln P)$$

ANOVA

	df	SS	MS	F	Significance F
Regression	1	5962.452138	5962.452138	218.5471333	6.94812E-20
Residual	50	1364.111267	27.28222534		
Total	51	7326.563405			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	41.21113731	0.98981223	41.63530826	1.77275E-40	39.22304094	43.19923369
ln(p)	-31.90779974	2.158362071	-14.78333972	6.94812E-20	-36.24299754	-27.57260193

3 LOG-LIN

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.917273231
R Square	0.84139018
Adjusted R Square	0.838217984
Standard Error	0.159901558
Observations	52

$$\ln q = f(P)$$

ANOVA

	df	SS	MS	F	Significance F
Regression	1	6.781765427	6.781765427	265.2389939	1.24301E-21
Residual	50	1.278425417	0.025568508		
Total	51	8.060190844			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	4.330785337	0.063226834	68.49600201	4.25001E-51	4.203790504	4.45778017
p	-0.664223668	0.040784549	-16.28615958	1.24301E-21	-0.746141844	-0.582305491

4 LOG-LOG

Regression Statistics	
Multiple R	0.955844442
R Square	0.913638597
Adjusted R Square	0.911911369
Standard Error	0.117990626
Observations	52

$$\ln q = f(\ln P)$$

ANOVA

	df	SS	MS	F	Significance F
Regression	1	7.364101457	7.364101457	528.9623426	2.99987E-28
Residual	50	0.696089387	0.013921788		
Total	51	8.060190844			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	3.716943882	0.022359414	166.2361915	2.94446E-70	3.672033676	3.761854087
ln(p)	-1.121358001	0.048756431	-22.99918135	2.99987E-28	-1.219288176	-1.023427827

5 RECIPROCAL or INVERSE

<i>Regression Statistics</i>	
Multiple R	0.936498043
R Square	0.877028585
Adjusted R Square	0.874569157
Standard Error	4.244897807
Observations	52

$q = f(1/p)$	
q	p
31.24	1.45
Average	Average

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	6425.605535	6425.605535	356.5985576	2.10241E-24
Residual	50	900.9578697	18.01915739		
Total	51	7326.563405			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	-6.024364975	2.059200728	-2.925584133	0.005158384	-10.16039136	-1.88833859
1/p	48.36495894	2.561185488	18.88381735	2.10241E-24	43.22066649	53.50925139

Calculate the elasticity at the mean values of q and p.

Log-lin

$$\ln(q) = \beta_1 + \beta_2 p$$

3 LOG-LIN

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.917273231
R Square	0.84139018
Adjusted R Square	0.838217984
Standard Error	0.159901558
Observations	52

lnq = f(P)

ANOVA

	df	SS	MS	F	Significance F
Regression	1	6.781765427	6.781765427	265.2389939	1.24301E-21
Residual	50	1.278425417	0.025568508		
Total	51	8.060190844			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	4.330785337	0.063226834	68.49600201	4.25001E-51	4.203790504	4.45778017
p	-0.664223668	0.040784549	-16.28615958	1.24301E-21	-0.746141844	-0.582305491

$$\begin{aligned} p^* &= 1.3 \Rightarrow \ln Q^* = 4.33 + (-0.66) * 1.3 \\ &= 3.465 \\ \Rightarrow Q^* &= e^{3.465} \approx (2.37)^{3.465} = 31.98 \end{aligned}$$

Elasticity at p^* and Q^* : $\text{slope @ } (p^*, Q^*) * \frac{p^*}{Q^*}$

$$\ln Q = \beta_1 + \beta_2 p \Rightarrow \frac{\partial \ln Q}{\partial p} = \beta_2$$

$$\beta_2 = \frac{\partial \ln Q}{\partial p} = \underbrace{\frac{\partial \ln Q}{\partial Q} \frac{\partial Q}{\partial p}}_{\text{chain Rule}} = \frac{1}{Q} \frac{\partial Q}{\partial p}$$

$$\Rightarrow \frac{\partial Q}{\partial p} = \beta_2 Q^* \equiv \text{slope @ } p^*, Q^*$$

$$\text{Elasticity} = (-0.66 * 31/98) * \frac{1.3}{31/98} = \beta_2 * p^*$$

where the slope in the table come from?

$$\text{Model: } \ln Q = \beta_1 + \beta_2 P$$

$$\text{Slope definition} = \frac{dQ}{dp}$$

$$\text{we know based on model that } \beta_2 = \frac{d(\ln Q)}{dp} \quad (\star)$$

$$\frac{d \ln Q}{dp} = \frac{d \ln Q}{dQ} \frac{dQ}{dp} = \frac{1}{Q} \frac{dQ}{dp} \quad (\star\star)$$

β_2 slope

$$(\star\star) \Rightarrow \beta_2 = \frac{1}{Q} \frac{dQ}{dp} \Rightarrow \beta_2 Q = \frac{dQ}{dp}$$

$\underbrace{\hspace{10em}}_{\text{slope}}$

Model 4 $\ln Q = \beta_1 + \beta_2 \ln p$

$$\text{Elasticity @ } (p^*, Q^*) = \underbrace{\frac{dQ}{dp} \text{ @ } (p^*, Q^*)}_{\text{Slope @ } p^*, Q^*} \cdot \frac{p^*}{Q^*}$$

① $p^* = 1.31$

② Q^*

Reg Table

$$\ln Q^* = \beta_1 + \beta_2 \ln p^* \Rightarrow \ln Q^* = 3.72 + (-1.12) \cdot \ln 1.31$$

$$\Rightarrow \ln Q^* = 3.417$$

$$\Rightarrow Q^* = e^{3.417} \Rightarrow Q^* = 30.478$$

③ slope at p^*, Q^* : $\frac{dQ}{dp}$ @ (p^*, Q^*)

Based on model we know $\beta_2 = \frac{d \ln Q}{d \ln p}$

using the chain Rule we know $\frac{d \ln Q}{d \ln p} = \frac{\frac{d \ln Q}{d Q} \cdot dQ}{\frac{d \ln p}{dp} \cdot dp}$

$$\Rightarrow \frac{d \ln Q}{d \ln p} = \frac{\frac{1}{Q} dQ}{\frac{1}{p} dp} = \frac{dQ}{dp} \cdot \frac{p}{Q}$$

$$\Rightarrow \frac{dQ}{dp} = \frac{Q}{p} \underbrace{\frac{d \ln Q}{d \ln p}}_{\beta_2} = \boxed{\frac{Q}{p} \beta_2} \rightarrow \text{already in Table}$$

$$\text{slope @ } (p^*, Q^*) = (-1.12) * \frac{30.478}{1.31} = -26.06$$

④ Last step. Elasticity: slope * $\frac{p^*}{Q^*}$

$$-26.06 * \frac{1.31}{30.478} = -1.12$$